GEOLOGY • RESOURCE PROSPECTIVITY • NUCLEAR MATERIALS

URANIUM IN THE BRITISH ISLES

A geological assessment of the strongest documented natural occurrences - and why plutonium is a different question entirely.

KEY JUDGMENT

The strongest documented natural uranium mineralisation in the British Isles is in south-west England, especially the wider Cornwall–Devon granite and metalliferous vein province. The principal alternative is Caithness and Orkney, where uranium enrichment is documented in Devonian sedimentary rocks. Neither region is established as a commercially recoverable uranium resource.

> **A necessary distinction: plutonium**
> There is no meaningful geological target for plutonium. Any naturally occurring plutonium is present only in trace quantities, while practical inventories are associated with nuclear materials and reactor processes rather than mineable rock deposits.

Prepared 24 October 2026 | Evidence-led desk study using public geological, scientific and official sources.

Scope: Great Britain and Ireland; regional comparisons rather than site-level prospecting guidance. This is not a mineral resource estimate, radiological survey or investment recommendation.

01 / EXECUTIVE JUDGEMENT

Where is uranium most likely to occur?

The answer depends on whether “find” means a documented mineral occurrence, a broad geochemical anomaly, or a deposit large and consistent enough to mine. Public sources support two leading regions but do not support a confident claim that either contains an economic resource.

> **Best single answer for known natural mineralisation**
> South-west England (Cornwall–Devon) has the strongest historical evidence for discrete uranium-bearing veins and past extraction. For sediment-hosted uranium enrichment, Caithness–Orkney is the leading comparison.

Evidence-weighted regional comparison

REGION / MODEL	EVIDENCE IN PUBLIC RECORD	ASSESSMENT
South-west England Granite-linked polymetallic veins	Multiple historic uranium occurrences and small-scale production associated with the Cornubian granite system; South Terras is a prominent documented example. [1,2,3]	Rank 1 Strongest evidence for discrete natural mineralisation; economic scale unproven.
Caithness–Orkney Devonian sedimentary strata	Uranium-bearing phosphatic and carbonaceous horizons in the Orcadian Basin; historic exploration identified multiple anomalies. [1,4,5]	Rank 2 Strongest alternative; enrichment may be dispersed and low-grade.
Other Scotland Granite-related veins	Occurrences associated with the Helmsdale/Ousdale granite and Criffel–Dalbeattie granodiorite, plus a south-Scottish vein occurrence. [1,5]	Rank 3 Documented but more localised evidence.
Other England & Wales Shales and varied host rocks	Additional, generally local occurrences; BGS describes most as rare and usually below economic grade. [1]	Rank 4 Not better-supported than the two leading regions.
Ireland Broad regional context	The sources reviewed here do not establish an Irish province with evidence comparable to the top two regions. This is a limit of the reviewed record, not proof of absence.	Unranked Insufficient comparative evidence in this desk study.

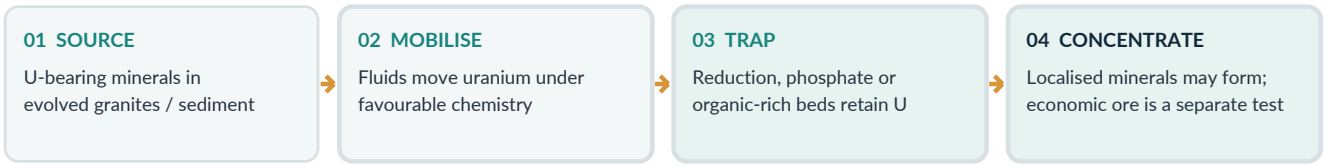
Important distinction

The region with the best evidence for uranium occurrence is not automatically the region with the best economic prospect. Resource size, continuity, grade, mineral processing, access, environmental controls and current appraisal all matter. The UK has no commercial uranium mining or milling according to the UK Radioactive Waste and Materials Inventory. [6]

02 / THE GEOLOGY BEHIND THE RANKING

How uranium becomes locally concentrated

Uranium is naturally present at low concentrations in many rocks. Its geology is governed by whether uranium can be released from source minerals, transported by fluids or groundwater, and then trapped in a smaller volume of rock. A radiometric or geochemical anomaly is evidence of enrichment, not by itself an orebody. [1,7]



Conceptual model only - not a prospecting map or resource estimate.

Two geological models dominate the British evidence

1. Granite-linked hydrothermal veins. Uranium can become enriched in evolved granitic systems and be redistributed during later fluid movement. Fractures and veins can host uranium minerals alongside tin, copper, lead, zinc, cobalt, nickel and other elements. In south-west England, BGS literature describes pitchblende-bearing veins within the wider metalliferous district. [1,2,3]

2. Sedimentary enrichment. Uranium can be retained by phosphate-rich material, organic matter and certain fine-grained sediments. The Caithness–Orkney Devonian successions contain phosphatic and carbonaceous horizons linked with uranium enrichment. Such enrichment can be spread through a substantial rock unit without forming a compact, mineable body. [1,4,5]

> **The concentration problem**
> A high reading in a specimen or a thin mineralised vein does not tell us the tonnage, continuity or recoverable quantity of uranium. Those require modern, representative geological appraisal and environmental assessment; they cannot be inferred from historic occurrences alone.

What the evidence can support

- The Cornubian granite province and the Orcadian Basin contain the clearest documented uranium-mineralisation settings in the UK literature.
- The mineralisation styles differ, so “best region” depends on whether the question concerns vein deposits or sediment-hosted enrichment.
- No public evidence reviewed for this report establishes an economically recoverable UK uranium resource.

03 / LEAD REGION

South-west England: Cornwall and Devon

Why it leads

The south-west has the clearest documented history of uranium as part of a broader mineralised granite province. The Cornubian batholith and its associated metalliferous veins are well known for tin and copper, with uranium minerals recorded in some of the later or cross-cutting mineralisation. The British Geological Survey describes uranium occurrences in this province as rare and commonly below economic grade, despite their well-established geological history. [1,2,3]

What the record shows

- Multiple confirmed occurrences. Uranium-bearing minerals have been recorded in Cornwall and Devon in association with polymetallic veins and altered rocks near granite intrusions. [1,2]
- Historic extraction. South Terras, near St Austell, is the best-known historic uranium mine in the UK literature. Other Cornish mines recorded small quantities or recovered uranium as a by-product of other metalliferous mining. [2,3]
- A plausible geological mechanism. Granite-related fluid systems and fractures provide a defensible explanation for local uranium concentration, rather than assuming that every granite is uranium-rich or that uranium is evenly distributed. [1,3]

Why this is not proof of a commercial deposit

The historical mineralised veins were irregular and localised; records of past production refer to limited historic output, not a modern reserve. Previous UK exploration campaigns did not establish significant economically recoverable uranium reserves. The distinction matters: a place can be the best documented location for natural uranium and still be a poor mining proposition. [2,3,4]

> **Assessment**
> Highest-confidence answer for documented natural uranium occurrences: the wider Cornwall–Devon metalliferous province. This ranking reflects historical occurrence evidence, not a prediction that an economic deposit exists or that a particular site should be explored.

Environmental note

Legacy uranium workings can leave uranium-bearing minerals and associated elements in spoil, soils or water. Research at South Terras has examined uranium and arsenic in mine-affected soils, demonstrating why legacy sites require radiological and environmental care rather than casual sampling. [3]

04 / MAIN ALTERNATIVE

Caithness and Orkney: sediment-hosted uranium

Different geology, similarly important evidence

Northern Scotland is the strongest alternative to south-west England, especially the Orcadian Basin's Devonian Old Red Sandstone succession. The Caithness flagstones and the sedimentary sequences across Orkney contain phosphatic and carbonaceous horizons where uranium enrichment is documented. The Geological Survey literature also records uranium-bearing minerals in sandstone units and nearby granite-related settings. [1,4,5]

Why this setting matters

- Large sedimentary basin. The Devonian sequence includes layered lake deposits and associated fine-grained sediments, creating varied chemical and depositional settings for uranium retention. [5]
- Phosphate and organic matter. BGS notes enrichment associated with phosphatic and black-shale horizons. These are plausible chemical sinks for uranium, but their presence does not establish a mineable resource. [1,5]
- Several styles of occurrence. The region includes sediment-hosted enrichment as well as granite-associated mineralisation, so anomalies do not all represent the same kind of deposit. [1,5]

The principal uncertainty

Sediment-hosted enrichment can be geographically broad yet comparatively dilute. The presence of uranium in phosphatic layers, black shales or groundwater may matter for geochemistry and environmental science without implying a concentrated orebody. A historic parliamentary answer identified Cornwall and the Orcadian rocks of Caithness, North Sutherland and Orkney as UK areas considered suitable for exploration, while explicitly stating that no economically proven deposits had been found at that time. [4]

> **Assessment**
> Best alternative for sediment-hosted uranium enrichment: Caithness–Orkney. This is a strong geological occurrence province, but public evidence does not support upgrading it to a commercial resource claim.

Additional northern Scottish evidence

The BGS also documents granite-related uranium veins around Ousdale/Helmsdale and in the Criffel–Dalbeattie region of southern Scotland. These support the wider pattern that uranium occurrences in Britain are associated with specific geological and structural settings, rather than evenly distributed across the landmass. [1,5]

05 / WIDER BRITISH ISLES

Other regions and the limits of comparison

South Scotland

The BGS regional account describes pitchblende in narrow polymetallic-carbonate breccia veins near the southern margin of the Criffel–Dalbeattie intrusive complex. This is a genuine occurrence, but the published description is of thin veins, not a regionally established uranium resource. [5]

Other parts of England and Wales

The BGS review notes additional occurrences, many associated with black shales. These are relevant to a nationwide geological picture, but the evidence reviewed here is less compelling as a broad, leading prospectivity case than either the Cornubian province or the Orcadian Basin. Presence of uranium in shale can reflect natural geochemical enrichment without the combination of grade, thickness and continuity needed for a mine. [1]

Ireland

Ireland has a diverse geological history, including Palaeozoic sedimentary basins, granites and mineralised belts. However, the reviewed public sources did not establish an Irish uranium province with evidence density or historical documentation comparable to south-west England or Caithness–Orkney. That is a scoped finding about the evidence reviewed here, not proof that no uranium occurrence exists in Ireland. A definitive all-island inventory would require systematic review of Geological Survey Ireland and GSNI occurrence datasets and technical reports. [8]

How to read this ranking

This is a qualitative comparison of published occurrence evidence, not a statistical model, a map of undiscovered resources or a claim that areas ranked lower are uranium-free. Exploration data are uneven in age, density and public availability; the UK's more systematic historical exploration largely took place decades ago. [4,6]

> **Regional conclusion**
> For an all-island desk study, start with the better-documented British provinces: south-west England for vein-hosted mineralisation and Caithness–Orkney for sediment-hosted enrichment. Treat the rest of the British Isles as a wider background, not as a proven ranking of commercial potential.

06 / ESSENTIAL DISTINCTION

Plutonium is not a geological prospect

Uranium is a naturally occurring element that can form mineral concentrations. Plutonium behaves differently: although trace natural occurrence is recognised, useful quantities are not found in ordinary geological deposits. The World Nuclear Association notes that, apart from trace quantities, plutonium is not now found in the Earth's crust. [9]

> **Direct answer to the plutonium part of the question**
> There is no "most likely place" in the British Isles to find a natural plutonium deposit. A geological prospectivity map for plutonium would be misleading because it implies a mineable natural resource where no such resource is established.

Where practical plutonium comes from

In practical terms, plutonium is generated when uranium-238 in nuclear fuel captures neutrons in a reactor. Plutonium recovered from irradiated fuel is handled through controlled nuclear-material processes; it is not obtained by mining a naturally concentrated plutonium ore. [9]

A separate source of trace quantities

Trace plutonium has also been dispersed in the environment by historical atmospheric weapons testing. That is an anthropogenic environmental legacy, not a natural geological deposit and not a reason to target any particular rock formation. [9]

Why the distinction matters

- A natural uranium occurrence says nothing about the presence of useful quantities of plutonium.
- No credible regional geological ranking for mineable plutonium can be produced from the British Isles' bedrock geology.
- For plutonium, the relevant subject is regulated nuclear-material management, not mineral exploration.

URANIUM	PLUTONIUM
Natural element, present in many rocks at low levels; locally concentrated by specific geological processes.	Trace natural occurrence only; meaningful inventories are associated with nuclear processes, not mineable geology.
Geology can identify natural mineralisation and geochemical anomalies.	Bedrock prospectivity is not a useful way to identify practical plutonium inventories.

07 / FINAL ASSESSMENT

The most defensible answer

If the question is "Where in the British Isles is uranium most likely to be found in naturally mineralised rock?", the strongest evidence-based answer is south-west England, especially the wider Cornwall–Devon granite and metalliferous vein province. If the question concerns the most notable alternative geological setting, it is Caithness–Orkney, where uranium is documented in Devonian sedimentary formations. [1–5]

That answer is about documented occurrence, not commercial viability. Historical exploration did not establish a significant economically recoverable UK uranium resource, and official UK inventory material states that there is no commercial uranium mining or milling. [4,6]

For plutonium, the answer is categorical: no British Isles geological region can responsibly be ranked as a place to find a mineable natural plutonium deposit. The useful quantities are associated with nuclear fuel and regulated materials management, not ordinary bedrock. [9]

Evidence gaps and confidence

- Geological occurrence confidence: high that the two leading regions contain documented uranium mineralisation; moderate for the exact comparative ranking because the styles differ.
- Economic-resource confidence: low. The reviewed public record does not establish a commercial resource, and mineral occurrences do not by themselves demonstrate tonnage or recoverability.
- Data freshness: much detailed UK uranium exploration cited in historical sources dates from the mid-twentieth century to the early 1980s. The BGS geogenic radionuclide synthesis is useful for regional context but is not a modern resource estimate. [1,4]
- Ireland coverage: limited in this scoped desk study. A stronger all-island statement would need a dedicated, systematic review of Irish and Northern Irish mineral occurrence records.

Responsible next step for academic research

Consult current British Geological Survey and Geological Survey Ireland mineral-occurrence and geochemical datasets, published regional geology, and official environmental/radiological records. Any field work near historic workings should be undertaken only with landowner permission, appropriate professional oversight and radiological/environmental safeguards. This report deliberately remains at regional scale and does not provide site-level prospecting directions.

> **Bottom line**
> Uranium: south-west England leads on documented vein occurrences; Caithness–Orkney leads among sediment-hosted settings. Plutonium: not a geological resource target.

08 / SOURCES AND METHOD

Selected references

The assessment prioritises British Geological Survey material, scientific literature and official public information. Sources are used to establish documented occurrence and geological setting; they do not all serve the same purpose, and none alone substitutes for a modern reserve estimate. Web references accessed 24 October 2026.

[1] Tye, A. M., Milodowski, A. E. & Smedley, P. L. (2017). Distribution of natural radioactivity in the environment: Geogenic radionuclides in British groundwater. British Geological Survey Internal Report OR/17/001.
https://earthwise.bgs.ac.uk/index.php/OR/17/001_Geogenic_radionuclides_in_British_groundwater
Regional synthesis of uranium occurrence in south-west England, Caithness–Orkney, granite-associated veins and other British settings.

[2] Corkhill, C. L. et al. (2017). Multi-scale investigation of uranium attenuation by arsenic at an abandoned uranium mine, South Terras. npj Materials Degradation, 1, 19.
https://www.nature.com/articles/s41529-017-0019-9
Peer-reviewed study of the historic South Terras uranium mine and the behaviour of uranium and arsenic in affected soils.

[3] Joint Nature Conservation Committee (GCR Series No. 36). Mineralization of England and Wales: South-west England chapter.
https://data.jncc.gov.uk/data/b007d9e3-a803-4464-bc61-85f4a1d5dd3f/gcr-v36-mineralization-england-wales-c7.pdf
Geological conservation account describing uranium-bearing veins and the regional Cornish metalliferous context.

[4] UK Parliament, Hansard (4 December 1980). Uranium Exploration, written answer.
https://hansard.parliament.uk/commons/1980-12-04/debates/5f19a139-2dc3-4bb0-9ed0-c7cc21f4c690/UraniumExploration
Historical statement naming Cornwall and the Orcadian rocks of Caithness, North Sutherland and Orkney as exploration prospects, while noting no economically proven deposits at that time.

[5] British Geological Survey. South of Scotland British Regional Geology; and The Northern Highlands of Scotland British Regional Geology.
https://webapps.bgs.ac.uk/memoirs/docs/B07006.html
Regional descriptions of the Criffel–Dalbeattie and Helmsdale/Ousdale uranium occurrences and of the northern Scottish sedimentary context. Also see BGS Northern Highlands of Scotland regional geology.

[6] UK Radioactive Waste and Materials Inventory / Nuclear Decommissioning Authority. How is radioactive waste produced? - Uranium mining and milling.
https://ukinventory.nda.gov.uk/information-hub/about-radioactive-waste/how-is-radioactive-waste-produced/
Official UK information stating that known UK uranium deposits are generally small and uneconomic and that there is no commercial mining or milling.

[7] MineralsUK / British Geological Survey (2010). Commodity Profile: Uranium.
https://nora.nerc.ac.uk/id/eprint/534458/1/uranium_profile.pdf
General explanation of uranium geochemistry and historic UK occurrences and exploration campaigns. Some resource statistics in this older profile are historical and are not treated here as current.

[8] Geological Survey Ireland. Minerals data and maps.
https://www.gsi.ie/data-and-maps/minerals/
Official portal for Irish mineral occurrence data; included as a relevant source for a fuller all-island review.

[9] World Nuclear Association (updated 30 March 2026). Plutonium.
https://world-nuclear.org/information-library/nuclear-fuel-cycle/fuel-recycling/plutonium
Explains that plutonium is produced in reactors and that natural plutonium is present, at most, in trace quantities in the Earth's crust.

[10] International Atomic Energy Agency. Uranium and Thorium Deposits and Occurrences (UThDEPO).
https://infcis.iaea.org/UTHDEPO
International database context; its documentation explicitly cautions that recorded deposits and occurrences do not necessarily imply commercial or economic extraction.

Method and caveats. This is a source-based regional synthesis, not a new geochemical survey. It does not estimate probabilities of discovery, quantify a uranium resource, inspect confidential exploration records, or assert that every area has been exhaustively assessed. Historical locality and deposit evidence is used only to compare geological settings at regional scale.

Prepared for informational and educational use. Geological and radiological conditions vary locally; never collect or handle radioactive minerals or legacy mine waste without appropriate expertise and permission.